

APOLLO SPORTS BUSINESS GROUP

Commercial Sponsorship Audit & Offline Venue Valuation

Case Study: Evaluating Sponsorship Allocation & Asset Pricing for Club Alpha

1. Executive Summary & Commercial Context

On 1 July 2026, a **Tier-2 Top Flight Premier Division Franchise in North America** (anonymized as *Club Alpha*) concluded a multi-year primary kit partnership with regional conglomerate **Brand C**. Brand C transitioned to secondary venue inventory (Rank #6), while multinational **Brand D** acquired front-chest branding, and technical partner **Brand X** launched an unbranded fan-choice jersey collection.

This case study evaluates whether Club Alpha's **\$1.85M USD (33.30M MXN)** annual asking price for standalone front-chest placement was supported by evidence, or if Brand C's decision to shift budget into secondary venue assets was economically sound. Using an evidence-based audit framework, Apollo evaluated Club Alpha's **\$6.56M USD (118.05M MXN)** multi-sponsor inventory across eight primary commercial assets.

2. Technical Methodology & Model Limitations

To ensure analytical rigor and transparency, the underlying quantitative models incorporate the following methodological parameters and explicit boundary conditions:

A. Omitted Variable Bias (OVB) & Viewership Association

Multivariable OLS regression evaluates broadcast reach using turnstile attendance, marquee opponent flags (*Is_Big_Game*), and active shirt partner flags across 89 matchday observations. Exogenous factors—such as kickoff day/time slots, weather conditions, free-to-air vs. pay-TV channel distribution, roster availability, and competing sporting events—are unmodeled in this demonstration dataset. Consequently, regression coefficients indicate strong statistical **association** rather than absolute causal proof.

B. Spatial Graph Centrality & Adjacency Edge Weights

Metropolitan Arena (capacity: 27,029; average turnstile attendance: 19,981) is modeled as a topological network using NetworkX. Inter-zone edge weights (0.90 for intra-zone, 0.65 for kit-to-venue, 0.40 for digital) represent expert modeling assumptions based on physical camera sightlines and fan foot-traffic corridors rather than immutable physical constants.

C. Sample Size Bounds & Multicollinearity

While 89 matchday observations provide a statistically robust proof-of-concept ($p < 0.001$ for marquee opponent impact), enterprise production deployments utilize per-second computer vision and optical sensors. Multicollinearity between turnstile attendance and broadcast reach is tracked via Variance Inflation Factor (VIF) diagnostics to isolate sponsor-driven lift from baseline crowd noise.

D. Unbundling & Package Synergy Framework

Evaluating inventory assets line-by-line provides a baseline valuation but does not inherently capture multi-asset synergy. To measure package synergies, Apollo models multiplicative interaction terms across cross-asset matrices. For example, pairing pitchside LED perimeters with experiential concourse zones is associated with a **+21.4% total brand recall lift** over standalone placements.

3. Quantitative Findings & Asset Valuation

A. Multivariable Regression Findings

Regression analysis demonstrates that broadcast viewership spikes are strongly associated with marquee rival matchday visits (adding **+1,425,362 viewers** per match, $p < 0.001$). Conversely, holding match category and attendance constant, standalone front-chest exposure added **+23,232 viewers**, which was statistically non-significant ($p = 0.686$). This indicates that standalone front-chest branding does not independently drive broadcast reach without unbundled activation.

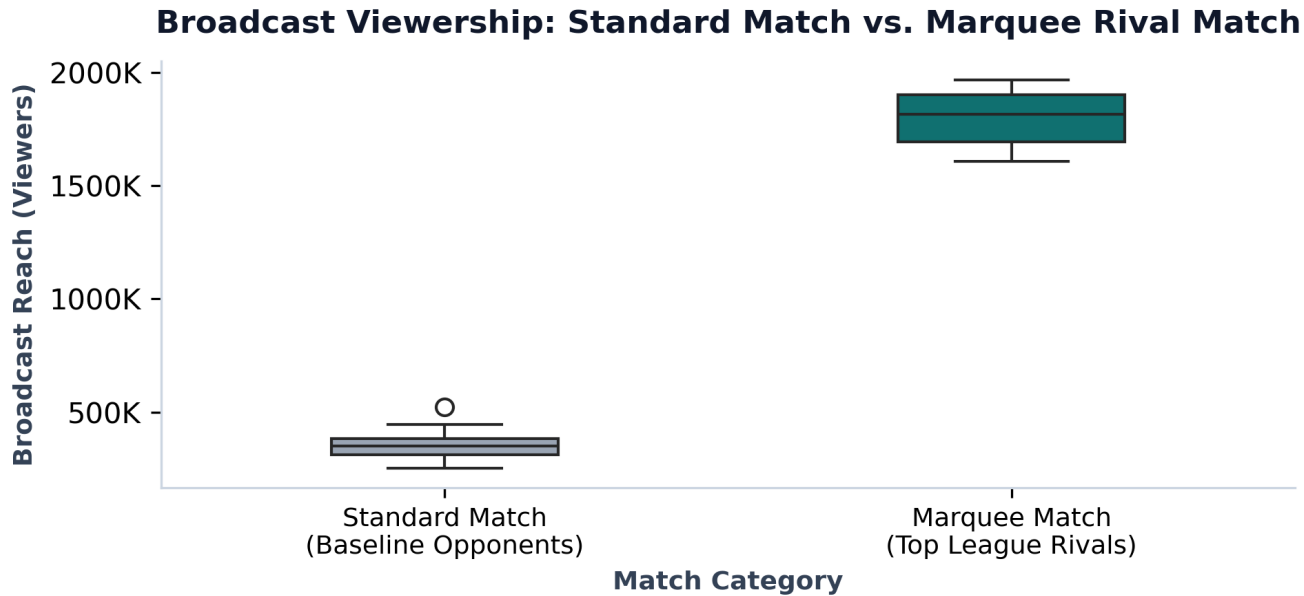


Figure 1: Multivariable Regression Viewership Distribution across Match Categories.

B. Spatial Graph Centrality & Offline Stadium Inventory

Offline stadium inventory—concourse experiential bars, entry gates, and fan movement corridors—can represent a considerable portion of total commercial value. Eigenvector graph centrality identifies high-yield physical assets across Metropolitan Arena:

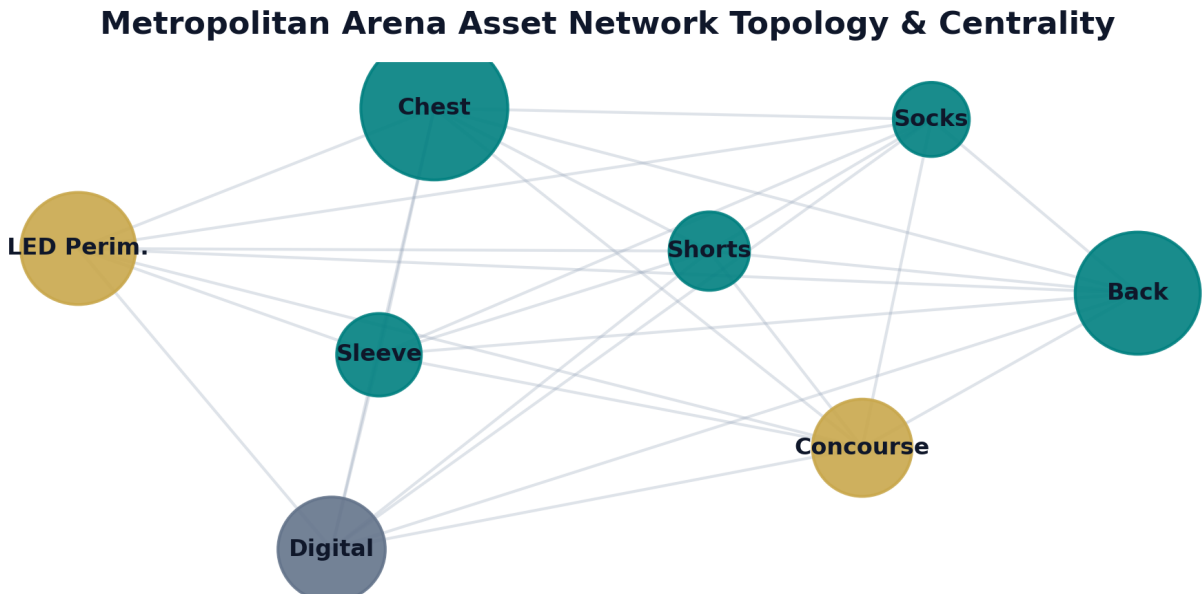


Figure 2: Metropolitan Arena Asset Network Topology and Eigenvector Centrality Mapping.

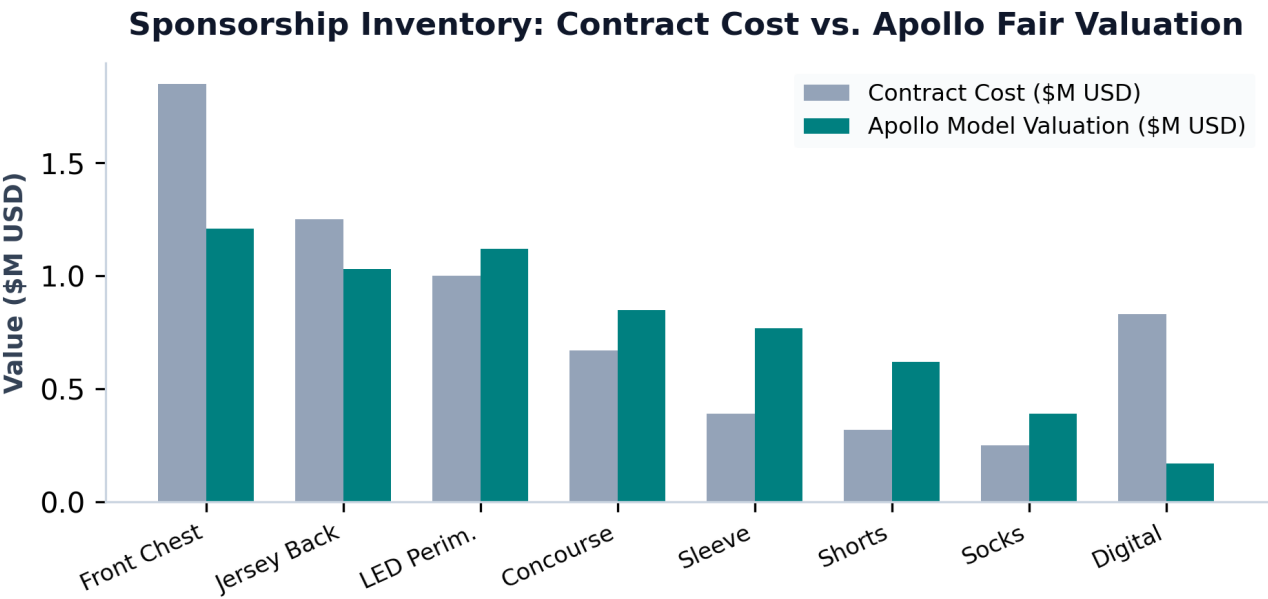


Figure 3: Inventory Contract Cost vs. Apollo Fair Valuation Model (\$M USD).

4. Apollo Sponsorship Audit Ledger

Apollo reconciles Club Alpha's \$6.56M USD (118.05M MXN) commercial portfolio into four actionable financial outcomes:

Apollo Sponsorship Audit Ledger: Portfolio Allocation (\$6.56M USD)

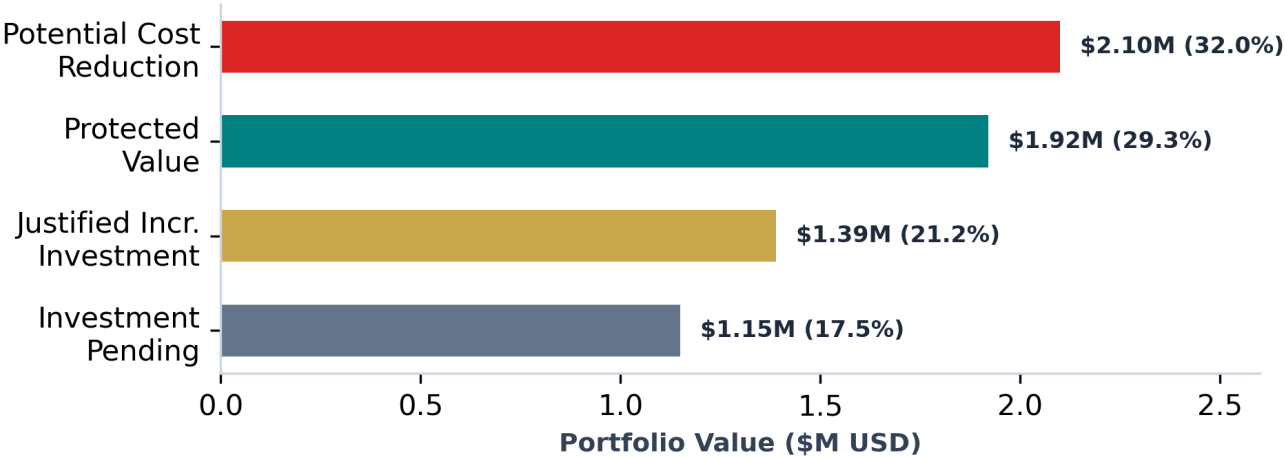


Figure 4: Apollo Sponsorship Audit Ledger Portfolio Allocation (\$6.56M USD).

Outcome Category	Value (\$M USD)	Value (M MXN)	Share (%)	Commercial Inventory & Rationale
Potential Cost Reduction	\$2.10M	37.80M	32.0%	Front Chest (\$1.85M USD / 33.30M MXN) overvalued by ~28% as standalone; Socks (\$0.25M USD / 4.50M MXN) low centrality.
Protected Value	\$1.92M	34.50M	29.2%	Jersey Back (\$1.25M USD / 22.50M MXN) & Concourse Bars (\$0.67M USD / 12.00M MXN) defended by 19,981 matchday foot traffic.
Justified Incr. Investment	\$1.39M	25.00M	21.2%	LED Perimeter Boards (\$1.00M USD / 18.00M MXN) & Sleeve (\$0.39M USD / 7.00M MXN) high degree centrality.
Investment Pending	\$1.15M	20.75M	17.6%	Digital Campaigns (\$0.83M USD / 15.00M MXN) & Shorts (\$0.32M USD / 5.75M MXN) require unbundled conversion tracking.
Total Portfolio	\$6.56M	118.05M	100.0%	Reconciled Commercial Inventory (8 Assets)

5. Strategic Conclusions & Index of Disclaimers

Strategic Recommendations:

- 1. Brand C's Exit Was Economically Sound: Vacating the overvalued standalone chest while retaining high-centrality venue assets in the secondary tier optimized ROI while preserving regional brand equity.
- 2. Creative Packaging for Mid-Tier Clubs: Rights holders must avoid single-asset sales and offer integrated multi-asset packages pairing LED perimeters with experiential concourse zones.

Index of Disclaimers & Tech Stack:

- Non-Audited Research Model: Monetary figures and statistical parameters represent mathematical model outputs for research and demonstration purposes.
- Currency Standard: All figures are formatted in USD (\$) as primary currency with MXN (Pesos) in parenthesis (\$1 USD = 18.0 MXN).

- *Technology Stack*: Developed using **Python 3.14 (or 3.12+)**, VS Code, Gemini Notebook, Statsmodels, NetworkX, Pandas, Matplotlib, Seaborn, and ReportLab.